INSTITUTE SEMINAR

Time-Periodic Compressible Navier–Stokes Equations in Besov Spaces

Periodic maximal regularity and low/high frequency analysis



Jiten Kumbhakar
Joint work with Kuntal Bhandari, Thomas Eiter,
Šárka Nečasová, and Yoshihiro Shibata
Institute of Mathematics, Czech Academy of Sciences, Prague

20 October 2026, 9:00 AM

© 2026 Jiten Kumbhakar. All rights reserved.

The periodic compressible-flow problem

Barotropic flow on $\mathbb{T} \times \mathbb{R}^N$, $N \geq 3$

$$\begin{cases} \partial_t \rho + \operatorname{div}(\rho u) = 0, \\ \rho[\partial_t u + (u \cdot \nabla)u] - \mu \Delta u - (\mu + \mu')\nabla \operatorname{div} u + \nabla P(\rho) = \rho h. \end{cases}$$

$$\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}, \qquad h(t + 2\pi, x) = h(t, x).$$

Equilibrium and physical assumptions

$$\rho_* > 0, \qquad P'(\rho_*) > 0,$$

$$\mu > 0, \qquad \frac{2\mu}{N} + \mu' \geq 0.$$

- Smooth P near the state $(\rho_*, 0)$.

The periodic response

- A small force with period 2π.
- A small strong solution with the same period.
- Estimates for the density and velocity.

© 2026 Jiten Kumbhakar. All rights reserved.

Periodic solutions through a semigroup

An autonomous model

$$Z'(t) = AZ(t) + H(t), \qquad H(t+2\pi) = H(t), \qquad S(t) = e^{tA}.$$

$$Z(t) = S(t)Z_0 + \int_0^t S(t-\tau)H(\tau)\,d\tau.$$

- Periodicity requires $Z(2\pi) = Z_0$, hence

$$(I - S(2\pi))Z_0 = \int_0^{2\pi} S(2\pi-\tau)H(\tau)\,d\tau.$$

- If $\|S(2\pi)\| < 1$, the Neumann series converges:

$$(I - S(2\pi))^{-1} = \sum_{n=0}^{\infty} S(2\pi)^n.$$

- This initial value and uniqueness give $Z(t+2\pi) = Z(t)$.

This model will later be replaced by a non-autonomous evolution family.

© 2026 Jiten Kumbhakar. All rights reserved.

Selected time-periodic compressible-flow results

Work	Setting	Result or method
Valli (1983)	Bounded domain	Small periodic solutions via stability.
Feireisl–Matušů-Nečasová–Petzeltová–Straškraba (1999)	Barotropic flow	Weak time-periodic solutions.
Ma–Ukai–Yang (2010)	$\mathbb{R}^N$, $N \geq 5$	Energy estimates and decay.
Feireisl–Mucha–Novotný–Pokorný (2012)	Full NSF system	Weak periodic solutions with thermal dissipation.
Kagei–Tsuda (2015)	$\mathbb{R}^N$, $N \geq 3$	Odd force, weighted Sobolev spaces.
Jin–Yang (2015); Jin (2016)	$\mathbb{R}^3$; $\mathbb{R}^4$	Small strong periodic solutions.
Tsuda (2016)	$\mathbb{R}^N$, $N \geq 3$	General small forcing and stability.

Our objective: periodic maximal regularity in spatial hybrid Besov spaces and time-periodic Besov spaces.

NSF denotes Navier–Stokes–Fourier.

© 2026 Jiten Kumbhakar. All rights reserved.

An equivalent system near equilibrium

$$\varphi = \frac{\rho - \rho_*}{\rho_*}, \quad w = \frac{u}{\gamma}, \quad \gamma = \sqrt{P'(\rho_*)}, \quad \nu = \frac{\mu}{\rho_*}, \quad \tilde{\nu} = \frac{\mu + \mu'}{\rho_*}.$$

$$\begin{cases} \partial_t \varphi + \gamma \operatorname{div} w = \widetilde{F}(\varphi, w), \\ \partial_t w - \nu \Delta w - \tilde{\nu} \nabla \operatorname{div} w + \gamma \nabla \varphi = \widetilde{G}(\varphi, w). \end{cases}$$

$$\widetilde{F}(\varphi, w) = -\gamma \operatorname{div}(\varphi w),$$

$$\widetilde{G}(\varphi, w) = -\varphi \partial_t w - \gamma(1+\varphi)(w \cdot \nabla) w - \nabla(\widetilde{P}(\varphi)\varphi^2) + \frac{1+\varphi}{\gamma} h,$$

$$\widetilde{P}(\varphi) = \frac{\rho_*}{\gamma} \int_0^1 (1-\theta) P''(\rho_*(1+\theta\varphi))\, d\theta.$$

- Recover $\rho = \rho_*(1+\varphi)$ and $u = \gamma w$.
- If $\|\varphi\|_\infty \leq \frac{1}{2}$, then $\rho \geq \rho_*/2$: the two periodic problems are equivalent.

© 2026 Jiten Kumbhakar. All rights reserved.

Kagei–Tsuda and our approach

	Kagei–Tsuda (2015)	Our approach
Forcing	Small, spatially odd, with spatial weights.	Small, with no spatial oddness or zero-time-mean assumption.
Spatial setting	Weighted Sobolev spaces.	Homogeneous Besov spaces with separate low and high frequencies.
Low frequencies	Spectral analysis of the time-T map.	Stationary problem, resolvents, and periodic multipliers.
High frequencies	Weighted energy estimates and evolution over one period.	Effective velocity, dyadic energy, and a non-autonomous evolution family.

- The estimates will determine the regularity required in each frequency region.
- Tsuda (2016) already treats general small forcing. Our focus is the regularity framework.

Kagei–Tsuda, JDE 258 (2015), 399–444, doi:10.1016/j.jde.2014.09.016. Tsuda, ARMA 219 (2016), 637–678, doi:10.1007/s00205-015-0902-x.

© 2026 Jiten Kumbhakar. All rights reserved.

Recall: homogeneous Besov spaces

$$\|u\|_{\dot{B}^s_{q,r}} = \big\|(2^{js}\|\dot{\Delta}_j u\|_{L^q})_{j\in\mathbb{Z}}\big\|_{\ell^r}, \qquad u \in \mathcal{S}'_h(\mathbb{R}^N).$$

$$s < \frac{N}{q}, \qquad \text{or} \qquad s = \frac{N}{q},\ r = 1 \quad \text{(the Banach-space realization used here).}$$

- Spatial differentiation shifts the regularity index:

$$\|\nabla u\|_{\dot{B}^{s-1}_{q,r}} \asymp \|u\|_{\dot{B}^s_{q,r}}.$$

- The endpoint summation gives boundedness:

$$\dot{B}^{N/q}_{q,1} \hookrightarrow L^\infty, \qquad \|u\|_\infty \lesssim \sum_j 2^{jN/q}\|\dot{\Delta}_j u\|_{L^q}.$$

- If $s_1 < s_2$, there is no general inclusion $\dot{B}^{s_2}_{q,r} \hookrightarrow \dot{B}^{s_1}_{q,r}$ on $\mathbb{R}^N$.

Bahouri–Chemin–Danchin (2011), Chapter 2.

© 2026 Jiten Kumbhakar. All rights reserved.

Transport as a source loses one derivative

- Suppose $\varphi \in \dot{B}^{s}_{q,r}$ and $w \in \dot{B}^{N/q}_{q,1}$.
- Apply the product law at regularity $s-1$:

$$\begin{aligned}\|w \cdot \nabla\varphi\|_{\dot{B}^{s-1}_{q,r}} &\lesssim \|w\|_{\dot{B}^{N/q}_{q,1}} \|\nabla\varphi\|_{\dot{B}^{s-1}_{q,r}} \\ &\lesssim \|w\|_{\dot{B}^{N/q}_{q,1}} \|\varphi\|_{\dot{B}^{s}_{q,r}}.\end{aligned}$$

- The density has index s, but this source has index $s-1$.
- The leading interaction makes the missing derivative visible:

$$2^{js}\|\dot{S}_{j-1}w \cdot \nabla\dot{\Delta}_j\varphi\|_{L^q} \lesssim \|w\|_{\infty} \underbrace{2^{j}}_{\text{derivative cost}} (2^{js}\|\dot{\Delta}_j\varphi\|_{L^q}).$$

$-\min(N/q, N/q') < s-1 < N/q$, $1 < q < \infty$, $q' = q/(q-1)$. The high–low interaction prevents a general same-index source bound.

© 2026 Jiten Kumbhakar. All rights reserved.

Hybrid homogeneous Besov spaces

$$u^\ell = \sum_{j<j_0} \dot{\Delta}_j u, \qquad u^h = \sum_{j\geq j_0} \dot{\Delta}_j u, \qquad u = u^\ell + u^h.$$

$$\boxed{\|u\|_{\dot{B}^{s_1,s_2}_{q_1,q_2,r}} = \|u^\ell\|_{\dot{B}^{s_1}_{q_1,r}} + \|u^h\|_{\dot{B}^{s_2}_{q_2,r}}.}$$

Low frequencies: $j < j_0$

$$\|u^\ell\|_{\dot{B}^{s+\delta}_{q,r}} \lesssim 2^{\delta j_0}\|u^\ell\|_{\dot{B}^{s}_{q,r}}.$$

- An extra derivative costs a fixed factor.

High frequencies: $j \geq j_0$

$$\|u^h\|_{\dot{B}^{s}_{q,r}} \lesssim 2^{-\delta j_0}\|u^h\|_{\dot{B}^{s+\delta}_{q,r}}.$$

- Higher regularity controls lower regularity.

$$\dot{B}^{N/q_1,N/q_2}_{q_1,q_2,1} \hookrightarrow L^\infty, \qquad \|uv\|_{\dot{B}^{s_1,s_2}_{q_1,q_2,r}} \lesssim \|u\|_{\dot{B}^{s_1,s_2}_{q_1,q_2,r}}\|v\|_{\dot{B}^{N/q_1,N/q_2}_{q_1,q_2,1}}.$$

$\delta > 0$. Product estimate in the admissible multiplier range. The cutoff j_0 is fixed and later chosen sufficiently large.

© 2026 Jiten Kumbhakar. All rights reserved.

Transport in the low and high frequency regions

- Let $\widetilde{w}$ be a prescribed transport field. The product law first controls $\widetilde{w} \cdot \nabla\varphi$ one spatial level below φ.

Low frequencies recover the index

$$\|(\widetilde{w} \cdot \nabla\varphi)^{\ell}\|_{\dot{B}^{s_1+1}_{q_1,r}} \lesssim 2^{j_0} \|(\widetilde{w} \cdot \nabla\varphi)^{\ell}\|_{\dot{B}^{s_1}_{q_1,r}}.$$

- Apply the product estimate before this inclusion.
- Small transport can be restored as a perturbation.

High frequencies retain the loss

$$\dot{B}^{s_2}_{q_2,r} \not\hookrightarrow \dot{B}^{s_2+1}_{q_2,r} \quad \text{on high frequencies.}$$

- The factor 2^j remains unbounded.
- Keep transport in the equation:

$$\partial_t\varphi + \gamma\widetilde{w} \cdot \nabla\varphi.$$

The same derivative count leads to two different linear constructions.

Mixed low–high products remain part of the product estimate.

© 2026 Jiten Kumbhakar. All rights reserved.

Time Fourier series give resolvent equations

$$\partial_t U^\ell - A^\ell U^\ell = K^\ell, \qquad U^\ell = \begin{pmatrix} \varphi^\ell \\ w^\ell \end{pmatrix}, \quad K^\ell = \begin{pmatrix} H^\ell \\ G^\ell \end{pmatrix},$$

$$A^\ell \begin{pmatrix} \varphi^\ell \\ w^\ell \end{pmatrix} = \begin{pmatrix} -\gamma \operatorname{div} w^\ell \\ \nu \Delta w^\ell + \tilde{\nu} \nabla \operatorname{div} w^\ell - \gamma \nabla \varphi^\ell \end{pmatrix}.$$

- For either $V = U^\ell$ or $V = K^\ell$, define

$$\mathcal{F}_{\mathbb{T}}[V](k) = \frac{1}{2\pi} \int_0^{2\pi} e^{-ikt} V(t)\, dt.$$

- Integration by parts uses the periodic boundary values:

$$\mathcal{F}_{\mathbb{T}}[\partial_t U^\ell](k) = \underbrace{\frac{[e^{-ikt} U^\ell(t)]_0^{2\pi}}{2\pi}}_{=0} + ik\, \mathcal{F}_{\mathbb{T}}[U^\ell](k).$$

$$\boxed{(ik - A^\ell) \mathcal{F}_{\mathbb{T}}[U^\ell](k) = \mathcal{F}_{\mathbb{T}}[K^\ell](k).}$$

$k = 0$ is the stationary mode. The nonzero modes correspond to $\lambda = ik$.

© 2026 Jiten Kumbhakar. All rights reserved.

The constant-coefficient resolvent problem

$$\begin{cases} \lambda\varphi + \gamma \operatorname{div} w = f, \\ \lambda w - \nu\Delta w - \tilde{\nu}\nabla \operatorname{div} w + \gamma\nabla\varphi = g. \end{cases}$$

$$\varphi = \mathcal{F}^{-1}\left[\frac{(\lambda + (\nu+\tilde{\nu})|\xi|^2)\mathcal{F}[f](\xi) - \gamma i\xi \cdot \mathcal{F}[g](\xi)}{\lambda^2 + ((\nu+\tilde{\nu})\lambda + \gamma^2)|\xi|^2}\right].$$

$$w = \mathcal{F}^{-1}\left[\frac{P(\xi)\mathcal{F}[g](\xi)}{\lambda + \nu|\xi|^2} + \frac{\lambda Q(\xi)\mathcal{F}[g](\xi) - \gamma i\xi\,\mathcal{F}[f](\xi)}{\lambda^2 + ((\nu+\tilde{\nu})\lambda + \gamma^2)|\xi|^2}\right].$$

$$Q(\xi) = \frac{\xi \otimes \xi}{|\xi|^2}, \qquad P(\xi) = I - Q(\xi), \qquad \xi \neq 0.$$

- These are spatial Fourier multipliers, before dyadic localization.
- Apply the low-frequency cutoff to the data and the solution.

P is the Leray projector and $Q = I - P$.

© 2026 Jiten Kumbhakar. All rights reserved.

Mikhlin's Fourier multiplier theorem

Theorem 2.17 in the manuscript

Let $1 < p < \infty$ and $m \in C^k(\mathbb{R}^N \setminus \{0\})$, where $k > N/2$ is an integer. Assume, for every multi-index α with $|\alpha| \le k$,

$$\sup_{\xi \in \mathbb{R}^N \setminus \{0\}} |\xi|^{|\alpha|} \left|\partial^\alpha m(\xi)\right| < \infty.$$

Then

$$T_m f = \mathcal{F}^{-1}[m(\xi)\mathcal{F}[f](\xi)] \quad \text{extends to a bounded operator on } L^p(\mathbb{R}^N),$$

with

$$\|T_m f\|_{L^p} \le C \|f\|_{L^p}.$$

- We must verify these derivative bounds for the resolvent symbols.
- Uniform symbol bounds give estimates uniform in the parameter λ.

Grafakos, *Classical Fourier Analysis*, 3rd ed., Springer (2014), Theorem 5.2.7. The symbol k here is the differentiability order in the theorem.

© 2026 Jiten Kumbhakar. All rights reserved.

The sector and low-frequency resolvent bounds

$$\Sigma_{\epsilon,\lambda_0} = \{\lambda \neq 0 : |\arg\lambda| \leq \pi - \epsilon,\ |\lambda| \geq \lambda_0\}, \qquad \operatorname{Re}\lambda \geq 0, \quad 0 < \epsilon < \pi/2.$$

- The normalized low-frequency symbols satisfy Mikhlin bounds. For example,

$$\sup_{\lambda,\,\xi\neq 0} |\xi|^{|\alpha|} \left| \partial_\xi^\alpha \left[\chi(2^{-j_0}\xi) \frac{(1+|\lambda|)(\lambda + (\nu+\tilde{\nu})|\xi|^2)}{\lambda^2 + ((\nu+\tilde{\nu})\lambda + \gamma^2)|\xi|^2} \right] \right| \leq C_\alpha.$$

$$\begin{aligned}(1+|\lambda|)\|\varphi_\lambda^\ell\|_{\dot{B}^{s_1+1}_{q_1,r}} + |\lambda|\|w_\lambda^\ell\|_{\dot{B}^{s_1}_{q_1,r}} \\ + |\lambda|^{1/2}\|w_\lambda^\ell\|_{\dot{B}^{s_1+1}_{q_1,r}} + \|w_\lambda^\ell\|_{\dot{B}^{s_1+2}_{q_1,r}} \\ \leq C(\|f^\ell\|_{\dot{B}^{s_1+1}_{q_1,r}} + \|g^\ell\|_{\dot{B}^{s_1}_{q_1,r}}).\end{aligned}$$

- Write $\varphi_\lambda^\ell = \mathcal{S}(\lambda)(f^\ell, g^\ell)$ and $w_\lambda^\ell = \mathcal{T}(\lambda)(f^\ell, g^\ell)$.

$1 < q_1 < \infty$, $1 \leq r < \infty$, $s_1 < N/q_1 - 2$, or equality with $r = 1$. The same block estimates control the critical components. For $\lambda = ik \neq 0$, choose $\lambda_0 \leq 1$. Details in B1.

© 2026 Jiten Kumbhakar. All rights reserved.

The missing temporal mode: the stationary problem

$$\gamma \operatorname{div} w^\ell = f^\ell, \qquad -\nu\Delta w^\ell - \tilde{\nu}\nabla \operatorname{div} w^\ell + \gamma\nabla\varphi^\ell = g^\ell.$$

$$\varphi_0^\ell = \frac{\nu+\tilde{\nu}}{\gamma^2} f^\ell - \frac{1}{\gamma}(-\Delta)^{-1} \operatorname{div} g^\ell,$$

$$w_0^\ell = \frac{1}{\nu}(-\Delta)^{-1} P g^\ell - \frac{1}{\gamma}\nabla(-\Delta)^{-1} f^\ell.$$

$$\|\varphi_0^\ell\|_{\dot{B}^{s_1+1}_{q_1,r}} + \|w_0^\ell\|_{\dot{B}^{s_1+2}_{q_1,r}} \lesssim \|f^\ell\|_{\dot{B}^{s_1+1}_{q_1,r}} + \|g^\ell\|_{\dot{B}^{s_1}_{q_1,r}}.$$

- Homogeneous derivative equivalence recovers the density and velocity norms from $\nabla\varphi_0^\ell$ and $\nabla^2 w_0^\ell$.
- These derivative bounds alone do not control the undifferentiated low-frequency block of the corresponding inhomogeneous norm.
- The realization in $\mathcal{S}_h'$ removes the polynomial ambiguity.

Define $\mathcal{S}(0)$ and $\mathcal{T}(0)$ by these stationary solution operators.

© 2026 Jiten Kumbhakar. All rights reserved.

The periodic reconstruction is a multiplier problem

$$K^\ell = (H^\ell, G^\ell), \qquad U^\ell = (\varphi^\ell, w^\ell).$$

$$M_k = \begin{cases} (\mathcal{S}(ik), \mathcal{T}(ik)), & k \neq 0, \\ (\mathcal{S}(0), \mathcal{T}(0)), & k = 0, \end{cases}$$

$$\boxed{U^\ell = \mathcal{F}_{\mathbb{T}}^{-1}\left[M_k \, \mathcal{F}_{\mathbb{T}}[K^\ell](k)\right].}$$

- Solving each Fourier mode gives this candidate solution.
- A multiplier theorem must control the reconstructed function in time.
- The spatial norms already contain critical ℓ^1 Besov components, and our index range includes $r = 1$.
- We first compare with the standard UMD-based multiplier route.

The zero temporal mode uses the stationary estimate, not a limit of the nonzero-mode bound.

© 2026 Jiten Kumbhakar. All rights reserved.

Weis's multiplier theorem and its hypotheses

- **UMD**: unconditional martingale differences; equivalently, boundedness of the Hilbert transform on $L^p(\mathbb{R}; X)$, $1 < p < \infty$.
- R-**boundedness**: uniform control of randomized sums,

$$\left(\mathbb{E} \left\| \sum_j \varepsilon_j T_j x_j \right\|_Y^2 \right)^{1/2} \leq C \left(\mathbb{E} \left\| \sum_j \varepsilon_j x_j \right\|_X^2 \right)^{1/2},$$

where $\varepsilon_j = \pm 1$ are independent Rademacher signs.

Weis: an operator-valued Mikhlin theorem

For UMD X, Y and $M \in C^1(\mathbb{R} \setminus \{0\}; \mathcal{L}(X, Y))$, if

$$\{M(\tau) : \tau \neq 0\}, \qquad \{\tau M'(\tau) : \tau \neq 0\}$$

are R-bounded, then M is a multiplier from $L^p(\mathbb{R}; X)$ to $L^p(\mathbb{R}; Y)$, $1 < p < \infty$.

Critical $\dot{B}^{N/q}_{q,1}$ is non-UMD. This route does not cover the full framework, including $r = 1$.

Weis, Math. Ann. 319 (2001), 735–758, Theorem 3.4, doi:10.1007/PL00004457. Some low-frequency realizations with $r > 1$ can be UMD.

© 2026 Jiten Kumbhakar. All rights reserved.

Besov spaces on the time torus

An inhomogeneous decomposition in time

$$\widetilde{\Delta}_j u = \mathcal{F}_{\mathbb{T}}^{-1}\left[\widetilde{\psi}_j(k)\mathcal{F}_{\mathbb{T}}[u](k)\right], \qquad u = \sum_{j\geq 0}\widetilde{\Delta}_j u.$$

$$\boxed{\|u\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};X)} = \left\|\left(2^{j\tilde{s}}\|\widetilde{\Delta}_j u\|_{L^{\tilde{q}}(\mathbb{T};X)}\right)_{j\geq 0}\right\|_{\ell^{\tilde{r}}}.}$$

- The temporal low block retains the stationary mode.
- For $\tilde{s} > 1/\tilde{q}$, or $\tilde{s} = 1/\tilde{q}$ and $\tilde{r} = 1$,

$$B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};X) \hookrightarrow C(\mathbb{T};X) \hookrightarrow L^{\infty}(\mathbb{T};X).$$

- A bounded bilinear map $B : X \times Y \to Z$ lifts to

$$\|B(u,v)\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};Z)} \lesssim \|u\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};X)}\|v\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};Y)}.$$

The time blocks are inhomogeneous, while the spatial blocks are homogeneous.

© 2026 Jiten Kumbhakar. All rights reserved.

The Arendt–Bu periodic multiplier theorem

Arbitrary Banach spaces X and Y

Let $(M_k)_{k\in\mathbb{Z}} \subset \mathcal{L}(X,Y)$ satisfy

$$\sup_{k\in\mathbb{Z}} \|M_k\|_{\mathcal{L}(X,Y)} < \infty,$$

$$\sup_{k\in\mathbb{Z}} \|k(M_{k+1} - M_k)\|_{\mathcal{L}(X,Y)} < \infty,$$

$$\sup_{k\in\mathbb{Z}} \|k^2(M_{k+1} - 2M_k + M_{k-1})\|_{\mathcal{L}(X,Y)} < \infty.$$

Then, for $\tilde{s} \in \mathbb{R}$ and $1 \le \tilde{q}, \tilde{r} \le \infty$,

$$\|\mathcal{F}_{\mathbb{T}}^{-1}[M_k \mathcal{F}_{\mathbb{T}}[f](k)]\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};Y)} \le C\|f\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};X)}.$$

- Ordinary operator norms suffice under these difference conditions.
- No UMD or reflexivity assumption is required.

Arendt–Bu, Proc. Edinburgh Math. Soc. 47 (2004), 15–33, Theorem 4.5(i), doi:10.1017/S0013091502000378.

© 2026 Jiten Kumbhakar. All rights reserved.

Resolvent identities verify the difference bounds

- For nonzero adjacent modes, $M_k = (ik - A^\ell)^{-1}$ gives

$$M_{k+1} - M_k = -iM_{k+1}M_k,$$
$$M_{k+1} - 2M_k + M_{k-1} = -2M_{k+1}M_kM_{k-1}.$$

- The resolvent estimate supplies a uniform solution bound and a $1/|k|$ bound at the forcing-space level.
- Therefore, for $|k| \geq 2$, the first and second differences have orders $|k|^{-1}$ and $|k|^{-2}$ in the required operator norm.
- The stationary bound controls the remaining finitely many differences directly, including those touching $k = 0$.

All three hypotheses of Arendt–Bu now follow from the resolvent and stationary estimates.

Complete first- and second-difference calculations are in backups B2–B3.

© 2026 Jiten Kumbhakar. All rights reserved.

The low-frequency time-periodic solution

$$U^\ell = \mathcal{F}_{\mathbb{T}}^{-1}\left[M_k \mathcal{F}_{\mathbb{T}}[K^\ell](k)\right], \qquad \partial_t U^\ell = A^\ell U^\ell + K^\ell.$$

$$\begin{aligned}
&\|\varphi^\ell\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};\dot{B}^{s_1+1}_{q_1,r})} + \|w^\ell\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};\dot{B}^{s_1+2}_{q_1,r})} \\
&\quad + \|\partial_t\varphi^\ell\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};\dot{B}^{s_1+1}_{q_1,r})} + \|\partial_t w^\ell\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};\dot{B}^{s_1}_{q_1,r})} \\
&\quad \lesssim \|H^\ell\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};\dot{B}^{s_1+1}_{q_1,r})} + \|G^\ell\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};\dot{B}^{s_1}_{q_1,r})}.
\end{aligned}$$

- The solution includes both the stationary mean and the oscillations.
- Repeat the block estimates for the critical spatial components.
- The equation gives the time derivatives after reconstructing the solution.

Spatial restrictions as on slide 14. Temporal condition: $\tilde{s} > 1/\tilde{q}$, or equality with $\tilde{r} = 1$.

© 2026 Jiten Kumbhakar. All rights reserved.

Restoring transport at low frequencies

$$\begin{cases} \partial_t \varphi^\ell + \gamma \operatorname{div} w^\ell = H^\ell - \gamma(\widetilde{w} \cdot \nabla \varphi^\ell)^\ell, \\ \partial_t w^\ell - \nu \Delta w^\ell - \tilde{\nu} \nabla \operatorname{div} w^\ell + \gamma \nabla \varphi^\ell = G^\ell. \end{cases}$$

- The product estimate first gives one lower spatial level.
- The inclusion from slide 9 recovers the required source level:

$$\|(\widetilde{w} \cdot \nabla \varphi^\ell)^\ell\|_{\dot{B}^{s_1+1}_{q_1,r}} \lesssim 2^{j_0} \|(\widetilde{w} \cdot \nabla \varphi^\ell)^\ell\|_{\dot{B}^{s_1}_{q_1,r}}.$$

- Apply the constant-coefficient periodic solver to this right-hand side. Small $\widetilde{w}$ makes the resulting map contractive.

Low-frequency result. The transport-perturbed diagonal system has a periodic solution with the same estimate as slide 21, including the critical components.

The smallness threshold depends on the fixed cutoff j_0.

© 2026 Jiten Kumbhakar. All rights reserved.

High frequencies retain density transport

$$\begin{cases} \partial_t \varphi^h + \gamma \operatorname{div} w^h + \gamma(\widetilde{w} \cdot \nabla \varphi^h)^h = H^h, \\ \partial_t w^h - \nu \Delta w^h - \tilde{\nu} \nabla \operatorname{div} w^h + \gamma \nabla \varphi^h = G^h. \end{cases}$$

- As a source, transport loses one density derivative. The high-frequency embedding cannot recover it.
- Keep this term in the density equation. Its coefficient depends on time.
- Temporal Fourier transformation now couples the modes:

$$\mathcal{F}_{\mathbb{T}}[\widetilde{w} \cdot \nabla \varphi](k) = \sum_{m \in \mathbb{Z}} \mathcal{F}_{\mathbb{T}}[\widetilde{w}](k-m) \cdot \nabla \mathcal{F}_{\mathbb{T}}[\varphi](m).$$

The previous diagonal temporal multiplier construction no longer applies directly. We use energy estimates and a non-autonomous evolution family.

The convolution identity is first read for smooth functions and then in distributions.

© 2026 Jiten Kumbhakar. All rights reserved.

Periodicity for a non-autonomous evolution

$$Z'(t) = A(t)Z(t) + H(t), \quad A(t+2\pi) = A(t), \quad H(t+2\pi) = H(t).$$

$$Z(t) = U(t,0)Z_0 + \int_0^t U(t,\tau)H(\tau)\,d\tau.$$

- The condition $Z(2\pi) = Z_0$ becomes

$$\big(I - U(2\pi,0)\big)Z_0 = \int_0^{2\pi} U(2\pi,\tau)H(\tau)\,d\tau.$$

- If $\|U(2\pi,0)\| < 1$, the inverse is a Neumann series.

$$\boxed{Z_{0,\mathrm{per}} = \big(I - U(2\pi,0)\big)^{-1} \int_0^{2\pi} U(2\pi,\tau)H(\tau)\,d\tau.}$$

Decay of the powers of $U(2\pi,0)$ also suffices; slide 30 supplies this justification.

© 2026 Jiten Kumbhakar. All rights reserved.

The basic energy has no explicit density damping

$$\partial_t \varphi + \gamma \operatorname{div} w + \gamma \widetilde{w} \cdot \nabla \varphi = H, \qquad \partial_t w - \nu \Delta w - \tilde{\nu} \nabla \operatorname{div} w + \gamma \nabla \varphi = G.$$

- The L^2 calculation displays the obstruction. Test with φ and w:

$$\frac{1}{2}\frac{d}{dt}(\|\varphi\|_2^2 + \|w\|_2^2) + \nu\|\nabla w\|_2^2 + \tilde{\nu}\|\operatorname{div} w\|_2^2 = \frac{\gamma}{2}\int_{\mathbb{R}^N} (\operatorname{div} \widetilde{w})|\varphi|^2 \, dx + \int_{\mathbb{R}^N} (H\varphi + G \cdot w) \, dx.$$

- Integration by parts cancels the two acoustic coupling terms.
- Viscosity gives dissipation for w, but no explicit positive term in φ remains on the left.
- We need to reveal the density damping hidden in the coupled system.

The L^2 identity explains the idea. The actual high-frequency estimates use dyadic L^{q_2} inequalities.

© 2026 Jiten Kumbhakar. All rights reserved.

Effective velocity reveals density damping

$$v := \nabla(-\Delta)^{-1}(\beta\varphi - \operatorname{div} w), \qquad \operatorname{div} v = \operatorname{div} w - \beta\varphi.$$

- In the transformed velocity equation, the first-order coupling is

$$[\beta(\nu + \tilde{\nu}) - \gamma]\nabla\varphi. \qquad \text{Choose} \quad \boxed{\beta = \frac{\gamma}{\nu + \tilde{\nu}}}.$$

$$\begin{cases} \partial_t \varphi + \beta\gamma\varphi + \gamma \operatorname{div} v + \gamma\widetilde{w} \cdot \nabla\varphi = H, \\ \partial_t v - (\nu + \tilde{\nu})\nabla \operatorname{div} v - \beta\gamma v + \beta^2\gamma\nabla(-\Delta)^{-1}\varphi \\ \qquad\qquad + \beta\gamma\nabla(-\Delta)^{-1}(\widetilde{w} \cdot \nabla\varphi) = -\nabla(-\Delta)^{-1} \operatorname{div} G + \beta\nabla(-\Delta)^{-1} H, \\ \partial_t Pw - \nu\Delta Pw = PG. \end{cases}$$

- The density now has the positive damping coefficient $\beta\gamma$.
- Localizing adds the commutator $R_j = \gamma[\widetilde{w} \cdot \nabla, \dot{\Delta}_j]\varphi$.

We suppress the superscript h in the displayed transformed system.

© 2026 Jiten Kumbhakar. All rights reserved.

High-frequency decay and the a priori estimate

$$X_m^h = \big(\dot{B}^{s_2+m}_{q_2,r} \cap \dot{B}^{N/q_2-1+m}_{q_2,1}\big)^h, \quad m = 0,1,2.$$

Small transport and sufficiently large j_0

$$\|\varphi^h\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};X_1^h)} + \|w^h\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};X_2^h)} + \|\partial_t\varphi^h\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};X_1^h)} + \|\partial_t w^h\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};X_0^h)}$$
$$\le C\big(\|H^h\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};X_1^h)} + \|G^h\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};X_0^h)}\big).$$

$$E_h(t) \le C_0 e^{-c_0(t-\tau)} E_h(\tau), \quad E_h(t) \asymp \|\varphi^h(t)\|_{X_1^h} + \|w^h(t)\|_{X_2^h} \quad (H = G = 0).$$

- Record the intermediate velocity level explicitly:

$$\|w^h\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};\dot{B}^{s_2^2+1}_{q_2^2,r})} \lesssim 2^{-j_0}\|w^h\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};\dot{B}^{s_2^2+2}_{q_2^2,r})}.$$

- High frequencies give $c(\nu+\tilde{\nu})2^{2j} - \beta\gamma \ge c_1 2^{2j} > 0$.
- Toward low frequencies, $e^{-c2^{2j}(t-\tau)} \to 1$ as $j \to -\infty$: there is no uniform positive decay rate there.

$E_h = \|\nabla\varphi^h\|_{X_0^h} + \|\nabla^2 v^h\|_{X_0^h} + \|\nabla^2 P w^h\|_{X_0^h}$. The same inclusion gives the critical intermediate velocity level.

© 2026 Jiten Kumbhakar. All rights reserved.

Existence begins with a frozen resolvent

$$0 = t_0 < t_1 < \cdots < t_n = 2\pi, \qquad t_m = \frac{2\pi m}{n}, \qquad \widetilde{w}_m(x) = \widetilde{w}(t_m, x).$$

$$\begin{cases} \lambda\varphi + \gamma \operatorname{div} w + \gamma \widetilde{w}_m \cdot \nabla\varphi = f, \\ \lambda w - \nu\Delta w - \tilde{\nu}\nabla \operatorname{div} w + \gamma\nabla\varphi = g. \end{cases}$$

- Apply $\dot{\Delta}_j$, $j \geq j_0$. Freezing time retains the spatial transport and its commutator.
- At the dyadic level, use the paper's base space and domain:

$$\mathbb{X}_{q_2} = W^{1,q_2} \times L^{q_2}, \qquad \mathbb{D}_{q_2} = W^{1,q_2} \times W^{2,q_2}.$$

- Abstractly, $(\lambda - A_m)(\varphi, w) = (f, g)$. The large-real-parameter estimate controls the frozen solution uniformly in the frozen time.

The dyadic forcing includes $\gamma[\widetilde{w}_m \cdot \nabla, \dot{\Delta}_j]\varphi$. Full Besov regularity follows from the a priori estimate.

© 2026 Jiten Kumbhakar. All rights reserved.

Solving the frozen problem by continuity

- Fix a large real $\lambda > 0$ and a dyadic level. Multiply the frozen transport by $\theta \in [0, 1]$.
- Let $R_\theta(\lambda)$ denote the resulting left-hand-side operator and set

$$S = \{\theta \in [0, 1] : R_\theta(\lambda) : \mathbb{D}_{q_2} \to \mathbb{X}_{q_2} \text{ is an isomorphism}\}.$$

- **Start at 0.** The constant-coefficient resolvent gives $0 \in S$.
- **Openness.** For θ near $\theta_0 \in S$,

$$R_\theta = \big[I + (\theta - \theta_0) B R_{\theta_0}^{-1}\big] R_{\theta_0}.$$

  The bracket is invertible by a Neumann series. Here B is the transport contribution.
- **Closedness.** Uniform a priori bounds make solutions Cauchy as $\theta_n \to \theta$.

Connectedness gives $S = [0, 1]$. In particular, the frozen problem is solvable at $\theta = 1$.

The continuation is performed at the dyadic construction level; the a priori estimates recover the final regularity.

© 2026 Jiten Kumbhakar. All rights reserved.

The evolution family and the periodic solution

- Join the frozen semigroups, with $h_n = 2\pi/n$ and $t \in [t_m, t_{m+1})$:

$$U_n(t,0) = e^{(t-t_m)A(t_m)} e^{h_n A(t_{m-1})} \cdots e^{h_n A(t_0)}.$$

- Uniform stability and the limit $n \to \infty$ give $U(t,s)$ with

$$\|U(t,s)\| \le C e^{-c(t-s)}.$$

$$M = U(2\pi, 0), \qquad \|M^n\| = \|U(2\pi n, 0)\| \le C e^{-2\pi c n}.$$

$$Z^h_{0,\mathrm{per}} = (I-M)^{-1} \int_0^{2\pi} U(2\pi,\tau) H^h(\tau)\, d\tau, \qquad (I-M)^{-1} = \sum_{n=0}^{\infty} M^n.$$

- This initial value gives the 2π-periodic high-frequency solution.
- Apply temporal blocks, commutator bounds, and the a priori estimate to recover the full time-Besov regularity.

Periodicity of the coefficients gives $M^n = U(2\pi n, 0)$. Decay of these powers suffices even when $\|M\| \ge 1$.

© 2026 Jiten Kumbhakar. All rights reserved.

The full hybrid linear periodic result

$$\begin{cases} \partial_t \varphi + \gamma \operatorname{div} w + \gamma \widetilde{w} \cdot \nabla \varphi = F, \\ \partial_t w - \nu \Delta w - \tilde{\nu} \nabla \operatorname{div} w + \gamma \nabla \varphi = G. \end{cases}$$

$$X_0 = \dot{B}^{s_1, s_2}_{q_1, q_2, r} \cap \dot{B}^{N/q_1 - 1, N/q_2 - 1}_{q_1, q_2, 1},$$

$$X_1 = \dot{B}^{s_1+1, s_2+1}_{q_1, q_2, r} \cap \dot{B}^{N/q_1, N/q_2}_{q_1, q_2, 1},$$

$$X_2 = \dot{B}^{s_1+2, s_2+2}_{q_1, q_2, r} \cap \dot{B}^{N/q_1+1, N/q_2+1}_{q_1, q_2, 1}.$$

For small $\|\widetilde{w}\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T}; X_2)}$ and $(F, G) \in B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T}; X_1) \times B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T}; X_0)$,

$$\begin{aligned} &\|\varphi\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T}; X_1)} + \|w\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T}; X_2)} + \|\partial_t \varphi\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T}; X_1)} + \|\partial_t w\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T}; X_0)} \\ &\qquad \le C\big(\|F\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T}; X_1)} + \|G\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T}; X_0)}\big). \end{aligned}$$

- Join the diagonal solutions and absorb $(\widetilde{w} \cdot \nabla \varphi^h)^\ell$ and $(\widetilde{w} \cdot \nabla \varphi^\ell)^h$ using smallness.

Index restrictions are recalled in B5; $\tilde{s} > 1/\tilde{q}$, or equality with $\tilde{r} = 1$.

© 2026 Jiten Kumbhakar. All rights reserved.

First nonlinear estimate: $\varphi \operatorname{div} w$

- After retaining transport in the density equation, its nonlinear source is $-\gamma\varphi \operatorname{div} w$.

$$\begin{aligned}\|\varphi \operatorname{div} w\|_{\dot{B}^{s_1+1,s_2+1}_{q_1,q_2,r}} &\lesssim \|\varphi\|_{\dot{B}^{s_1+1,s_2+1}_{q_1,q_2,r}} \|w\|_{\dot{B}^{N/q_1+1,N/q_2+1}_{q_1,q_2,1}} \\ &\quad + \|\varphi\|_{\dot{B}^{N/q_1,N/q_2}_{q_1,q_2,1}} \|w\|_{\dot{B}^{s_1+2,s_2+2}_{q_1,q_2,r}}, \\ \|\varphi \operatorname{div} w\|_{\dot{B}^{N/q_1,N/q_2}_{q_1,q_2,1}} &\lesssim \|\varphi\|_{\dot{B}^{N/q_1,N/q_2}_{q_1,q_2,1}} \|w\|_{\dot{B}^{N/q_1+1,N/q_2+1}_{q_1,q_2,1}}.\end{aligned}$$

- The hybrid product law preserves the density source space X_1.
- The spatial derivative falls on velocity, with $\nabla : X_2 \to X_1$.
- Lift the bilinear estimate to the time-periodic Besov space:

$$\boxed{\|\varphi \operatorname{div} w\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};X_1)} \lesssim \|\varphi\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};X_1)} \|w\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};X_2)}.}$$

The product and temporal embedding restrictions remain in force.

© 2026 Jiten Kumbhakar. All rights reserved.

Second nonlinear estimate: $(w \cdot \nabla)w$

- Use the critical velocity norm as a multiplier and the intermediate spatial level for the differentiated factor.

$$\|(w \cdot \nabla)w\|_{\dot{B}^{s_1,s_2}_{q_1,q_2,r}} \lesssim \|w\|_{\dot{B}^{N/q_1,N/q_2}_{q_1,q_2,1}} \|w\|_{\dot{B}^{s_1+1,s_2+1}_{q_1,q_2,r}},$$

$$\|(w \cdot \nabla)w\|_{\dot{B}^{N/q_1-1,N/q_2-1}_{q_1,q_2,1}} \lesssim \|w\|^2_{\dot{B}^{N/q_1,N/q_2}_{q_1,q_2,1}}.$$

$$\|(w \cdot \nabla)w\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};X_0)} \lesssim \|w\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};\dot{B}^{N/q_1,N/q_2}_{q_1,q_2,1})} \Big(\|w\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};\dot{B}^{s_1+1,s_2+1}_{q_1,q_2,r})} + \|w\|_{B^{\widetilde{s}}_{\widetilde{q},\widetilde{r}}(\mathbb{T};\dot{B}^{N/q_1,N/q_2}_{q_1,q_2,1})} \Big).$$

- The full hybrid norms retain both frequency regions.
- The explicit $s_2 + 1$ bound on slide 27 supplies the high-frequency intermediate velocity control.

Spatial input: the multiplier product law, with s_i and $N/q_i - 1$ in its admissible range. Temporal input: the bilinear estimate on slide 18.

© 2026 Jiten Kumbhakar. All rights reserved.

Small time-periodic solutions

- $N \geq 3$, $1 < q_1, q_2 < N$, $1 < \tilde{q} < \infty$, $1 \leq r, \tilde{r} < \infty$.
- For $i = 1, 2$,

$$-\min\left\{\frac{N}{q_i}, \frac{N}{q_i'}\right\} < s_i \leq \frac{N}{q_i} - 2, \quad \text{with the upper endpoint allowed only for } r = 1.$$

- $\tilde{s} > 1/\tilde{q}$, or $\tilde{s} = 1/\tilde{q}$ and $\tilde{r} = 1$.

Existence and regularity of a periodic solution

For sufficiently small $h \in B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T}; X_0)$, the transformed CNS system has a small 2π-periodic solution satisfying

$$\|\varphi\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};X_1)} + \|w\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};X_2)} + \|\partial_t \varphi\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};X_1)} + \|\partial_t w\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};X_0)} \leq C\|h\|_{B^{\tilde{s}}_{\tilde{q},\tilde{r}}(\mathbb{T};X_0)}.$$

- The nonlinear iteration uses the transport-perturbed linear solver and the product estimates.
- Smallness gives $\rho = \rho_*(1 + \varphi) \geq \rho_*/2$ and $u = \gamma w$.

X_0, X_1, X_2 are defined on slide 31.

© 2026 Jiten Kumbhakar. All rights reserved.

JK

INSTITUTE SEMINAR

Thank you!

Questions & discussion



Jiten Kumbhakar

Institute of Mathematics, Czech Academy of Sciences, Prague

© 2026 Jiten Kumbhakar. All rights reserved.

20 October 2026

B1. Checking the low-frequency symbols

- On the support of $\chi(2^{-j_0}\xi)$, $|\xi| \lesssim 2^{j_0}$, and

$$\left|\lambda^2 + ((\nu+\tilde{\nu})\lambda + \gamma^2)|\xi|^2\right| \geq c(|\lambda|^2 + (|\lambda|+1)|\xi|^2),$$

$$\operatorname{Re}\lambda \geq 0, \qquad |\lambda| \geq \lambda_0 > 0, \qquad c = c(\lambda_0, j_0, \nu, \tilde{\nu}, \gamma) > 0.$$

  - Differentiate the quotient. Each derivative of the reciprocal denominator is a sum of products of denominator derivatives divided by higher powers of that denominator.
  - Multiplying by $|\xi|^{|\alpha|}$ absorbs the powers from these derivatives. Cutoff derivatives have bounded support at $|\xi| \asymp 2^{j_0}$.
  - Normalize each symbol by the weight of its target norm: $1+|\lambda|$, $|\lambda|$, $|\lambda|^{1/2}|\xi|$, or $|\xi|^2$.

- Mikhlin gives the uniform L^{q_1} block estimate. Multiply by 2^{js_1} and sum in ℓ^r, $j < j_0$. Repeat for the critical ℓ^1 component.

The lower bound is used on the fixed low-frequency support; its constant may depend on the cutoff.

© 2026 Jiten Kumbhakar. All rights reserved.

B2. The first resolvent difference

- For nonzero k, $M_k = (ik - A^\ell)^{-1}$. On the low-frequency data space E and solution space F, the resolvent estimates give

$$\|M_k\|_{\mathcal{L}(E,F)} \le C, \qquad |k|\|M_k\|_{\mathcal{L}(E,E)} \le C.$$

For $k, k+1 \neq 0$, the resolvent identity yields

$$M_{k+1} - M_k = -iM_{k+1}M_k.$$

Consequently,

$$|k|\|M_{k+1} - M_k\|_{\mathcal{L}(E,F)} \le \|M_{k+1}\|_{\mathcal{L}(E,F)}\, |k|\|M_k\|_{\mathcal{L}(E,E)} \le C.$$

$$E = (\dot{B}^{s_1+1}_{q_1,r})^\ell \times (\dot{B}^{s_1}_{q_1,r})^\ell, \qquad F = (\dot{B}^{s_1+1}_{q_1,r})^\ell \times (\dot{B}^{s_1+2}_{q_1,r})^\ell.$$

- Include the critical components by intersection. Differences involving M_0 are estimated directly from the stationary bound.

The two mapping norms distinguish the base resolvent bound from the regularity bound.

© 2026 Jiten Kumbhakar. All rights reserved.

B3. Second differences and time derivatives

For $k-1, k, k+1 \neq 0$, the resolvents commute and

$$M_{k+1} - 2M_k + M_{k-1} = -2M_{k+1}M_kM_{k-1}.$$

Thus, for $|k| \geq 2$,

$$k^2 \|M_{k+1} - 2M_k + M_{k-1}\|_{\mathcal{L}(E,F)} \leq C.$$

- The finite cases $k = -1, 0, 1$ use the stationary bound for M_0 and the nonzero-mode resolvent bounds.
- For time derivatives use the sequence $N_k = ikM_k$, with $N_0 = 0$, into the appropriate base component.

$$N_{k+1} - N_k = iM_{k+1} + ik(M_{k+1} - M_k),$$
$$N_{k+1} - 2N_k + N_{k-1} = ik(M_{k+1} - 2M_k + M_{k-1}) + i(M_{k+1} - M_{k-1}).$$

- The base resolvent bounds also verify the Arendt–Bu conditions for N_k.

No resolvent composition through the stationary operator is needed.

© 2026 Jiten Kumbhakar. All rights reserved.

B4. Why the high-frequency cutoff matters

$$v_j = \dot{\Delta}_j \nabla(-\Delta)^{-1}(\beta\varphi - \operatorname{div} w), \qquad \beta = \frac{\gamma}{\nu + \tilde{\nu}}.$$

$$c(\nu + \tilde{\nu})2^{2j} - \beta\gamma \geq c_1 2^{2j} \qquad (j \geq j_0 \text{ sufficiently large}).$$

- The effective density equation provides the damping $\beta\gamma\varphi_j$.
- The second-order part controls v_j and Pw_j. The cutoff absorbs lower-order coupling; transport commutators are controlled by smallness.
- Sum the weighted dyadic inequalities to control

$$E_h = \|\nabla\varphi^h\|_{X_0^h} + \|\nabla^2 v^h\|_{X_0^h} + \|\nabla^2 P w^h\|_{X_0^h}.$$

- Recover the intermediate level from $2^{j(s_2+1)} \leq 2^{-j_0} 2^{j(s_2+2)}$.
- Applying temporal blocks adds commutators with $\widetilde{w}$; the same smallness absorbs these terms and gives the time-Besov estimate.

The intermediate critical level follows by replacing s_2 with $N/q_2 - 1$.

© 2026 Jiten Kumbhakar. All rights reserved.

B5. Indices and the spaces in the final theorem

$$I_N(q,r) = \left\{ s : \; -\min\left(\frac{N}{q}, \frac{N}{q'}\right) < s \leq \frac{N}{q} - 2, \text{ with equality at the upper end only if } r = 1 \right\}.$$

$$X_0 = \dot{B}^{s_1,s_2}_{q_1,q_2,r} \cap \dot{B}^{N/q_1-1,N/q_2-1}_{q_1,q_2,1},$$
$$X_1 = \dot{B}^{s_1+1,s_2+1}_{q_1,q_2,r} \cap \dot{B}^{N/q_1,N/q_2}_{q_1,q_2,1},$$
$$X_2 = \dot{B}^{s_1+2,s_2+2}_{q_1,q_2,r} \cap \dot{B}^{N/q_1+1,N/q_2+1}_{q_1,q_2,1}.$$

- **Linear theorem:** $N \geq 1$, $1 < q_1, q_2, \tilde{q} < \infty$, $1 \leq r, \tilde{r} < \infty$, $s_i \in I_N(q_i, r)$.
- **Nonlinear theorem:** additionally $N \geq 3$ and $q_1, q_2 < N$.
- In both results, $\tilde{s} > 1/\tilde{q}$, or $\tilde{s} = 1/\tilde{q}$ and $\tilde{r} = 1$.

All spatial spaces are homogeneous; the temporal Besov space is inhomogeneous.

© 2026 Jiten Kumbhakar. All rights reserved.

B6. The frozen dyadic resolvent estimate

- For small frozen transport and $j \geq j_0$, Proposition 5.4 gives, uniformly in the frozen time,

$$\begin{aligned} &|\lambda|\|\dot{\Delta}_j\varphi\|_{L^{q_2}} + |\lambda|\|\nabla\dot{\Delta}_j\varphi\|_{L^{q_2}} + \|\nabla\dot{\Delta}_j\varphi\|_{L^{q_2}} \\ &\quad + |\lambda|\|\dot{\Delta}_j w\|_{L^{q_2}} + |\lambda|^{1/2}\|\nabla\dot{\Delta}_j w\|_{L^{q_2}} + \|\nabla^2\dot{\Delta}_j w\|_{L^{q_2}} \\ &\qquad \leq C(\|\dot{\Delta}_j f\|_{L^{q_2}} + \|\nabla\dot{\Delta}_j f\|_{L^{q_2}} + \|\dot{\Delta}_j g\|_{L^{q_2}}). \end{aligned}$$

$$\operatorname{Re}\lambda \geq 0, \quad |\lambda| \geq \lambda_0; \qquad \mathbb{X}_{q_2} = W^{1,q_2} \times L^{q_2}, \quad \mathbb{D}_{q_2} = W^{1,q_2} \times W^{2,q_2}.$$

- For large real $\lambda > 0$, this gives the base-space resolvent bound $\|(\lambda - A_m)^{-1}\| \lesssim 1/\lambda$.
- The continuation argument supplies solvability. Dyadic summation and the a priori estimate recover the full Besov regularity.

The forcing is understood with the localized transport commutator, as in the dyadic construction.

© 2026 Jiten Kumbhakar. All rights reserved.